

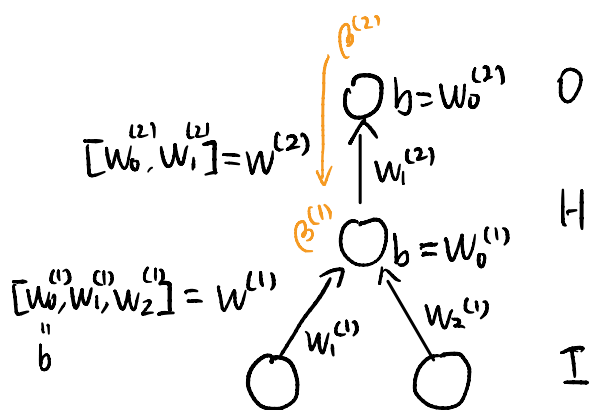
INT301 Bio-Computation

Week 5 Tutorial

For a feedforward neural network with two inputs and one output, use the back-propagation algorithm with momentum as shown below to update the weights after each of the training examples $\{(1, 0), 1\}$ and $\{(0, 1), 0\}$ in order (i.e. incremental learning version).

$$\Delta w(t) = -\frac{\partial E_e}{\partial w(t)} + \alpha \Delta w(t-1)$$

Assume the network has a single hidden layer with one neuron and all neurons use the sigmoid activation function. All weights (including bias) are initially equal to 1, learning rate $\eta = 0.2$ and momentum term $\alpha = 0.9$.



$$\begin{aligned} x &= (1, 0) \Rightarrow b = w_0 x_0 \\ x_1 \quad x_2 \quad x &= (x_0, x_1, x_2) \\ y &= 1 \\ \eta &= 0.2 \\ \alpha &= 0.9 \end{aligned}$$

Forward: $(I \rightarrow H)$

$$s^{(1)} = \sum w^{(1)} x = [1, 1, 1] \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 2$$

$$o^{(1)} = \sigma(s^{(1)}) = \sigma(2) = 0.8808$$

$(H \rightarrow O)$

$$\begin{aligned} s^{(2)} &= \sum w^{(2)} o = [w_0^{(2)}, w_1^{(2)}] \begin{bmatrix} o_0^{(1)} \\ o^{(1)} \end{bmatrix} \\ &= [1, 1] \begin{bmatrix} 1 \\ 0.8808 \end{bmatrix} = 1.8808 \end{aligned}$$

$$o^{(2)} = \sigma(s^{(2)}) = \sigma(1.8808) = 0.8677$$

$o_0^{(1)}$ 要考虑 bias

Backward: $\beta^{(2)} = o^{(2)} (1 - o^{(2)}) (y - o^{(2)})$
 $0 \rightarrow 1 \quad = 0.8677 (1 - 0.8677) (1 - 0.8677) = 0.0152$

$\Delta W^{(2)}$ $= \eta \beta^{(2)} o^{(1)} + \alpha \Delta W_{\text{previous}}^{(2)}$

$(\Delta W_{\text{previous}}^{(2)} = 0)$
 $= 0.2 \times 0.0152 [o_0^{(1)} \quad o^{(1)}] + \alpha \cdot 0 = 0.0027$
 bias 通常 0.8808
 - 总是 1

$\searrow = [w_0^{(2)}, w_1^{(2)}]$
 $= [0.0030, 0.0027]$

$\Delta W_0^{(2)}$ $= \eta \beta^{(2)} + \alpha \Delta W_{\text{previous}}^{(2)} = 0.2 \times 0.0152 + 0$
 $= 0.0030$

$H \rightarrow I: \beta^{(1)} = o^{(1)} (1 - o^{(1)}) \cdot \beta^{(2)} \cdot w_1^{(2)}$
 $= 0.8808 (1 - 0.8808) \cdot 0.0152 \cdot 1 = 0.0016$

$\Delta W^{(1)}$ $= \eta \beta^{(1)} \cdot x + \alpha \Delta W_{\text{previous}}^{(1)} = 0.2 \times 0.0016 \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}^T$
 $= [0.00032, 0.00032, 0]$ 3 weights

更新权重:

1st Data

$\Delta W_{\text{new}}^{(2)} = [w_0^{(2)}, w_1^{(2)}] + \Delta W^{(2)}$
 $= [1, 1] + [0.0030, 0.0027]$
 $= 1.0030 + 1.0027$

$\Delta W_{\text{new}}^{(1)} = [w_0^{(1)}, w_1^{(1)}, w_2^{(1)}] + \Delta W^{(1)}$
 $= [1, 1, 1] + [0.00032, 0.00032, 0]$
 $= [1.00032, 1.00032, 1]$

2nd Data:

$$X = (0, 1) \rightarrow X = [X_0, X_1, X_2]$$

$$y = 0 = [1, 0, 1]$$

$$W^{(2)} = [W_0^{(2)}, W_1^{(2)}] = [1.0030, 1.0027]$$

$$W^{(1)} = [W_0^{(1)}, W_1^{(1)}, W_2^{(1)}] = [1.00032, 1.00032, 1]$$

Forward:

$$S^{(1)} = \sum W^{(1)} X = 2.00032$$

$$O^{(1)} = G(S^{(1)}) = 0.8808 \quad I \rightarrow H$$

$$S^{(2)} = \sum W^{(2)} O = \sum W^{(2)} \begin{bmatrix} O_0^{(1)} & O^{(1)} \\ \parallel & \\ 1 & \end{bmatrix}$$

$$= [1.0030, 1.0027] [1, 0.8808]^T = 1.8862$$

$$O^{(2)} = G(S^{(2)}) = 0.8683$$

$$\text{Backward: } \beta^{(2)} = O^{(2)} (1 - O^{(2)}) (y - \underbrace{O^{(2)}}_0) = -0.0993$$

$O \rightarrow H$:

$$\Delta W^{(2)} = \eta \beta^{(2)} \begin{bmatrix} O_0^{(1)} \\ O^{(1)} \end{bmatrix} + \alpha \Delta W_{\text{previous}}^{(2)}$$

$$= 0.2 (-0.0993) \begin{bmatrix} 1 \\ 0.8808 \end{bmatrix}^T + 0.9 \cdot \begin{bmatrix} 0.0030 \\ 0.0027 \end{bmatrix}^T$$

$$= [-0.0172, -0.0151]$$

$$H \rightarrow I: \beta^{(1)} = O^{(1)} (1 - O^{(1)}) \beta^{(2)} W_1^{(2)}$$

$$= 0.8808 (1 - 0.8808) (-0.0993) 1.0027 = -0.0105$$

$$\Delta W^{(1)} = \eta \beta^{(1)} \alpha + \alpha \Delta W_{\text{previous}}^{(1)} = 0.2 (-0.0105) \begin{bmatrix} X_0 \\ X_1 \\ X_2 \end{bmatrix}^T + 0.9 \cdot \begin{bmatrix} 0.00032 \\ 0.00032 \\ 0 \end{bmatrix}^T$$

$$= \begin{bmatrix} -0.0008 \\ 0.0029 \\ -0.0021 \end{bmatrix}^T$$

更新

$$\begin{aligned}W_{\text{new}}^{(2)} &= [W_0^{(2)}, W_1^{(2)}] + \Delta W^{(2)} \\&= [1.0030, 1.0027] + [-0.0172, -0.0151] \\&= [0.9858, 0.9876]\end{aligned}$$

$$\begin{aligned}W_{\text{new}}^{(1)} &= [W_0^{(1)}, W_1^{(1)}, W_2^{(1)}] + \Delta W^{(1)} \\&= \quad \quad \quad + \begin{bmatrix} -0.0008 \\ 0.0029 \\ -0.0021 \end{bmatrix}^T \\&= [0.9985, 1.0006, 0.9979]\end{aligned}$$